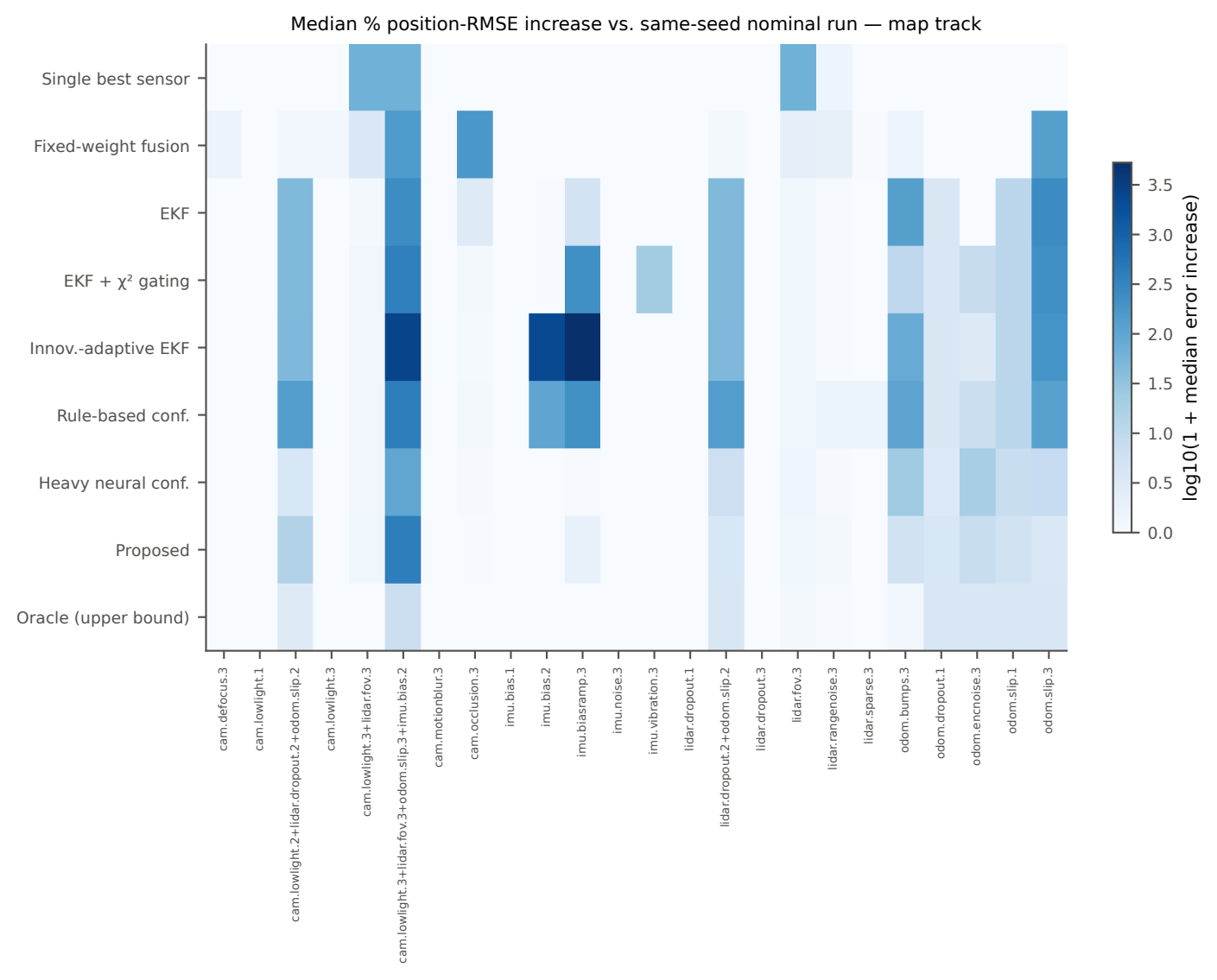
Median % position-RMSE increase vs. same-seed nominal run — map track
Single best sensor
Fixed-weight fusion
EKF
EKF + χ² gating
Innov.-adaptive EKF
Rule-based conf.
Heavy neural conf.
Proposed
Oracle (upper bound)
cam.defocus.3
cam.lowlight.1
cam.lowlight.2+lidar.dropout.2+odom.slip.2
cam.lowlight.3
cam.lowlight.3+lidar.fov.3
cam.lowlight.3+lidar.fov.3+odom.slip.3+imu.bias.2
cam.motionblur.3
cam.occlusion.3
imu.bias.1
imu.bias.2
imu.biasramp.3
imu.noise.3
imu.vibration.3
lidar.dropout.1
lidar.dropout.2+odom.slip.2
lidar.dropout.3
lidar.fov.3
lidar.rangenoise.3
lidar.sparse.3
odom.bumps.3
odom.dropout.1
odom.encnoise.3
odom.slip.1
odom.slip.3
log10(1 + median error increase)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5